

Stochastic Contextual Bandits / Part 1

Margaux Brégère

EDF R&D + LPSM, Sorbonne Université



Gilles Stoltz

Laboratoire de mathématiques d'Orsay



université
PARIS-SACLAY



K -armed stochastic bandits

Simplest possible framework

K probability distributions $\nu_1, \dots, \nu_K$ in a model $\mathcal{D}$
with expectations $\mu_1, \dots, \mu_K \longrightarrow \mu^\star = \max_{a \in [K]} \mu_a$

At each round $t = 1, 2, \dots$,

1. Statistician picks **arm** $A_t \in [K]$ (possibly at random)
2. She gets a reward Y_t drawn according to ν_{A_t}
3. This is the **only feedback** she receives

$\longrightarrow$ **Exploration–exploitation dilemma**
estimate the ν_a **vs.** get high rewards Y_t

Goal:

Maximize expected cumulative rewards $\longleftrightarrow$ Minimize **regret**

$$R_T = T\mu^\star - \mathbb{E}\left[\sum_{t=1}^T Y_t\right] = \sum_{a \in [K]} (\mu^\star - \mu_a) \mathbb{E}[N_a(T)]$$

$\longleftrightarrow$ **Sub-linear** control of the $\mathbb{E}[N_a(T)]$ for suboptimal arms a

Motivation(s)

One-armed bandits in casinos → K -armed stochastic bandits

Clinical trials? (cf. K treatments)

Online advertisement (cf. K ads and homogeneous population)

Optimization of a noisy environment

- Continuum of arms $\leftrightarrow$ optimization of a (regular) function
- Tree structure $\leftrightarrow$ chess, go

These lectures **targeted toward a specific application:**

Balance between demand and production in electricity systems

I.e., Brégère, Gaillard, Goude, Stoltz, ICML 2019

We will cover what is minimally required to do so

Excellent reference



Bandit Algorithms, Tor Lattimore and Csaba Szepesvári

Cambridge University Press, 2020

Setting of K -armed stochastic bandits:

Distributions $\nu_1, \dots, \nu_K$ in model $\mathcal{D}$ with expectations $\mu_1, \dots, \mu_K$

At each round $t \geq 1$, pick arm $A_t \in [K]$, get and observe $Y_t \sim \nu_{A_t}$

Proof of the rewriting of regret

Tower rule: $\mathbb{E}[Y_t | A_t] = \mu_{A_t}$ thus $\mathbb{E}[Y_t] = \mathbb{E}[\mu_{A_t}]$

$$\begin{aligned} R_T &= \sum_{t=1}^T (\mu^* - \mathbb{E}[Y_t]) = \sum_{t=1}^T (\mu^* - \mathbb{E}[\mu_{A_t}]) \\ &= \sum_{t=1}^T \sum_{a \in [K]} (\mu^* - \mu_a) \mathbb{E}[\mathbb{I}_{\{A_t=a\}}] = \sum_{a \in [K]} (\mu^* - \mu_a) \mathbb{E}[N_a(T)] \end{aligned}$$

where
$$N_a(T) = \sum_{t=1}^T \mathbb{I}_{\{A_t=a\}}$$

Noise assumption: σ^2 -sub-Gaussian

$$Y_t \sim \nu_{A_t} \quad \text{made more precise:} \quad Y_t = \mu_{A_t} + \eta_t$$

Noise η_t independent from “everything else” and σ^2 -sub-Gaussian:

$$\forall \lambda \in \mathbb{R}, \quad \mathbb{E}[e^{\lambda \eta_t} \mid \mathcal{F}_{t-1}] = \mathbb{E}[e^{\lambda \eta_t}] \leq e^{\lambda^2 \sigma^2 / 2}$$

$\mathcal{F}_{t-1}$ represents the information: $Y_1, \dots, Y_{t-1}$ and $A_1, \dots, A_t$

In particular, $\mathbb{E}[e^{\lambda \eta_t} \mid \mathcal{F}_{t-1}] = \mathbb{E}[e^{\lambda \eta_t}] = 0$ [proof by Taylor expansions]

Examples:

- $\mathcal{N}(0, \sigma^2)$ noise
- noise taking values in $[a, b]$ is $(b - a)^2/4$ -sub-Gaussian (cf. Hoeffding's lemma)

Explore-then-commit strategy

As a warm-up

Explore-then-commit strategy

Reference: e.g., Lattimore and Szepesvári [2020, Chapter 6]

1. Explore for mK rounds

Pull each arm m times, compute empirical means $\hat{\mu}_a$ for $a \in [K]$

2. Exploit for $T - mK$ rounds

For $t \geq mK + 1$, pull an arm $A_t \in \arg \max_{a \in [K]} \hat{\mu}_a$

Regret bound (under the σ^2 -sub-Gaussian noise assumption):

$$R_T \leq \underbrace{m \sum_{a \in [K]} (\mu^* - \mu_a)}_{\text{exploration cost}} + \underbrace{(T - mK) \sum_{a \in [K]} (\mu^* - \mu_a) \exp\left(-\frac{m(\mu^* - \mu_a)^2}{4\sigma^2}\right)}_{\text{exploitation cost}}$$

Hoeffding's inequality

Reminder: η is σ^2 -sG if $\mathbb{E}[e^{\lambda\eta}] \leq e^{\lambda^2\sigma^2/2}$

If η, η' are σ^2 -sG and independent, then $(\eta + \eta')/2$ is $\sigma^2/2$ -sG

Cramér-Chernoff: for $\lambda = \varepsilon/\sigma^2 > 0$,

$$\mathbb{P}(\eta \geq \varepsilon) \leq e^{-\lambda\varepsilon} \mathbb{E}[e^{\lambda\eta}] \leq \exp(-\lambda\varepsilon + \lambda^2\sigma^2/2) = \exp(-\varepsilon^2/(2\sigma^2))$$

Hoeffding's inequality: $\eta_1, \dots, \eta_m$ independent and σ^2 -sG

$$\mathbb{P}\left(\frac{1}{m} \sum_{t=1}^m \eta_t \geq \varepsilon\right) \leq \exp(-m\varepsilon^2/(2\sigma^2))$$

$$\text{or} \quad \text{w.p. } 1 - \delta, \quad \frac{1}{m} \sum_{t=1}^m \eta_t \leq \sigma \sqrt{\frac{2}{m} \ln(1/\delta)}$$

Back to explore-then-commit

Regret bound (with σ^2 -sub-Gaussian noises):

$$R_T = \sum_{t=1}^T (\mu^* - \mathbb{E}[\mu_{A_t}]) \leq \underbrace{m \sum_{a \in [K]} (\mu^* - \mu_a)}_{\text{exploration cost}} + \underbrace{(T - mK) \sum_{a \in [K]} (\mu^* - \mu_a) \exp\left(-\frac{m(\mu^* - \mu_a)^2}{4\sigma^2}\right)}_{\text{exploitation cost}}$$

Exploitation costs $\longleftrightarrow a^* \in \arg \max_{a \in [K]} \mu_a \neq \arg \max_{a \in [K]} \hat{\mu}_a$

$$\mathbb{P}(\hat{\mu}_a \geq \hat{\mu}_{a^*}) = \mathbb{P}\left(\frac{1}{m} \sum_{t=1}^m \underbrace{\eta_t - \eta'_t}_{2\sigma^2\text{-sG}} \geq \mu^* - \mu_a\right) \leq (-m\varepsilon^2/(4\sigma^2))$$

where $\varepsilon = \mu^* - \mu_a$

Two types of regret bounds

Gap-dependent regret bound:

$$R_T = \sum_{t=1}^T (\mu^* - \mathbb{E}[\mu_{A_t}]) \leq \underbrace{m \sum_{a \in [K]} (\mu^* - \mu_a)}_{\text{exploration cost}} + \underbrace{(T - mK) \sum_{a \in [K]} (\mu^* - \mu_a) \exp\left(-\frac{m(\mu^* - \mu_a)^2}{4\sigma^2}\right)}_{\text{exploitation cost}}$$

Involves the gaps $\Delta_a = \mu^* - \mu_a$

More generally called: problem-dependent bound

Now, uniform bound (**gap-free bound** or worst-case bound):

in a model $\mathcal{D}_{\sigma^2, B}$ with $\Delta_a \leq B$

Using $xe^{-x^2} \leq 1/2$ for $x \geq 0$

$$\sup_{\mathcal{D}_{\sigma^2, B}} R_T \leq mKB + (T - mK) \frac{2\sigma}{\sqrt{m}} \frac{1}{2} \leq mKB + \sigma \frac{T}{\sqrt{m}} = \mathcal{O}(T^{2/3})$$

for $m \sim T^{2/3}$

Upper-confidence-bound [UCB] strategy

“Optimism in the face of uncertainty”

Model: $\nu_1, \dots, \nu_K$ are σ^2 -sG

A popular strategy: **UCB** [upper confidence bound]

Auer, Cesa-Bianchi and Fisher [2002]

Pull each arm once

For $t \geq K$, pick $A_{t+1} \in \arg \max_{a \in [K]} \left\{ \hat{\mu}_a(t) + \sigma \sqrt{\frac{8 \ln t}{N_a(t)}} \right\}$

Exploitation: cf. empirical mean $\hat{\mu}_a(t) = \frac{1}{N_a(t)} \sum_{s=1}^t Y_s \mathbb{I}_{\{A_s=a\}}$

Exploration: cf. $\sigma \sqrt{8 \ln t / N_a(t)}$ favors arms a not pulled often

Regret bounds (slightly suboptimal) of two types

– **Gap-dependent** bound: $R_T \lesssim \square \sum_{a: \mu_a < \mu^*} \frac{\sigma \ln T}{\mu^* - \mu_a}$

– **Gap-free** bound: $\sup \sigma R_T \lesssim \square \sigma \sqrt{KT \ln T}$
 $\sigma^2\text{-sG}$

Proof of $R_T \lesssim \square \sum_{a: \mu_a < \mu^*} \frac{\sigma \ln T}{\mu^* - \mu_a}$

Hoeffding–Azuma ext.: $\mathbb{P} \left(|\mu_a - \hat{\mu}_a(t)| \leq \sigma \sqrt{\frac{8 \ln t}{N_a(t)}} \right) \geq 1 - 2t^{-3}$

Recall $\hat{\mu}_a(t) = \frac{1}{N_a(t)} \sum_{s=1}^t Y_s \mathbb{I}_{\{A_s=a\}}$

Indeed, by **Doob's optional skipping**:

$$\begin{aligned}
 & \mathbb{P} \left(|\mu_a - \hat{\mu}_a(t)| > \sigma \sqrt{\frac{8 \ln t}{N_a(t)}} \right) \\
 &= \sum_{n=1}^t \mathbb{P} \left(|\mu_a - \hat{\mu}_{a,n}| > \sigma \sqrt{\frac{8 \ln t}{n}} \quad \text{and} \quad N_a(t) = n \right) \\
 &\leq \underbrace{\sum_{n=1}^t \mathbb{P} \left(|\mu_a - \hat{\mu}_{a,n}| > \sigma \sqrt{\frac{2 \ln(1/t^{-4})}{n}} \right)}_{\leq 2t^{-4}, \quad \text{cf. classic Hoeffding-Azuma applied twice}}
 \end{aligned}$$

where $\hat{\mu}_{a,n}$ denotes a n -sample average with distribution ν_a

Proof of $R_T \lesssim \square \sum_{a: \mu_a < \mu^*} \frac{\sigma \ln T}{\mu^* - \mu_a}$

Hoeffding–Azuma ext.: $\mathbb{P}\left(|\mu_a - \hat{\mu}_a(t)| \leq \sigma \sqrt{\frac{8 \ln t}{N_a(t)}}\right) \geq 1 - 2t^{-3}$

If $A_t = b$ is not an optimal arm a^* , then by design

$$\hat{\mu}_{a^*}(t) + \sigma \sqrt{\frac{8 \ln t}{N_{a^*}(t)}} \leq \hat{\mu}_b(t) + \sigma \sqrt{\frac{8 \ln t}{N_b(t)}}$$

$$\text{thus w.p.} \geq 1 - 4t^{-3}, \quad \mu^* \leq \mu_b + 2\sigma \sqrt{\frac{8 \ln t}{N_b(t)}}$$

which imposes
$$N_b(t) \leq \frac{32\sigma \ln T}{(\mu^* - \mu_b)^2}$$

$\mathbb{E}[N_b(T)]$ controlled by this $+ \square t^{-2}$

Conclude with
$$R_T = \sum_{a \in [K]} (\mu^* - \mu_a) \mathbb{E}[N_a(T)]$$

Proof of $\sup_{\sigma^2\text{-sG}} R_T \lesssim \sigma \sqrt{KT \ln T}$

We proved $\mathbb{E}[N_b(t)] \lesssim \square \sigma \frac{\ln T}{(\mu^* - \mu_b)^2}$

$$\begin{aligned} \text{Thus } R_T &= \sum_{a \in [K]} (\mu^* - \mu_a) \sqrt{\mathbb{E}[N_a(T)]} \sqrt{\mathbb{E}[N_a(T)]} \\ &\leq \square \sigma \sqrt{\ln T} \sum_{a \in [K]} \sqrt{\mathbb{E}[N_a(T)]} \\ &\leq \square \sigma \sqrt{KT \ln T} \end{aligned}$$

cf. Jensen for $\sqrt{\cdot}$

Thompson sampling

A Bayesian view

Thompson sampling: a Bayesian view

Studied mostly for exponential families $\mathcal{D} \leftrightarrow \Theta$

E.g., Bernoulli $\Theta = (0, 1)$; Gaussian $\Theta = \mathbb{R}$ with known variance σ^2

Ref.: Thompson [1933]; Agrawal and Goyal [2012, 2013, 2017];

Kaufmann, Korda and Munos [2012, 2013]

Well-chosen prior distribution π over Θ

$\mathcal{D}$ Bernoulli $\rightarrow \pi$ Beta; $\mathcal{D}$ Gaussian $\rightarrow \pi$ Gaussian

Posterior distributions: $\hat{\pi}_{t,a}$ for arm $a \in [K]$

Based on π and rewards Y_s such that $1 \leq s \leq t-1$ and $A_s = a$

Strategy: draw $\theta_{t,a} \sim \hat{\pi}_{t,a}$ ^{exploration} and pick $A_t \in \arg \max_{a \in [K]} \mu(\theta_{t,a})$ ^{exploitation}

Thompson sampling: Bernoulli case $\Theta = (0, 1)$

Distributions indexed by $p_1, \dots, p_K$ with best parameter $p^\star$

Prior $\pi = \text{Beta}(1, 1) = \text{uniform distribution over } (0, 1)$

Posteriors: $\hat{\pi}_{t,a} = \text{Beta}(1 + S_a(t), 1 + N_a(t) - S_a(t))$

$$\text{where} \quad N_a(t) = \sum_{s \leq t} \mathbb{I}_{\{A_s = a\}} \quad \text{and} \quad S_a(t) = \sum_{s \leq t} Y_s \mathbb{I}_{\{A_s = a\}}$$

Regret bounds (complex proofs)

$$R_T \lesssim \sum_{a: p_a < p^\star} \frac{\ln T}{\text{kl}(p_a, p^\star)} \leq \sum_{a: p_a < p^\star} \frac{\ln T}{2(p^\star - p_a)^2}$$

thus also $R_T \lesssim \square \sqrt{KT \ln T}$

Optimal regret bounds

Achieved with more complex strategies

Optimal **distribution-dependent** bound

For all **reasonable** strategies, $\forall \nu_1, \dots, \nu_K$ in $\mathcal{D}$,

$$\liminf \frac{R_T}{\ln T} \geq \sum_{k \in [K]} \frac{\mu^* - \mu_k}{\mathcal{K}_{\text{inf}}(\nu_k, \mu^*)} \quad \text{where}$$

$$\mathcal{K}_{\text{inf}}(\nu_k, \mu^*) = \inf \left\{ \text{KL}(\nu_k, \xi) : \xi \in \mathcal{D}, \nu_k \ll \xi \text{ and } \mathbb{E}(\xi) > \mu^* \right\}$$

References: Lai et Robbins '85, Burnetas et Katehakis '96, Ménard et al. '19

Note: “ $\forall \nu_1, \dots, \nu_K$ in $\mathcal{D}$ ” **after** “for all [...] strategies”

Bound achieved by IMED and KL-UCB strategies:
models $\mathcal{D}$ given by 1-dim. exponential families or by $\mathcal{P}([0, 1])$

IMED: Honda and Takemura, series of papers in the 2010s

kl-UCB and KL-UCB: Garivier, Cappé, Ménard, Maillard, Munos, Hadiji and Stoltz,
series of papers in the 2010s

Optimal **distribution-free** bound for models

$$\mathcal{D} \supseteq \mathcal{Ber} = \{\text{Ber}(p) : p \in (0, 1)\}$$

For **all strategies**,

$$\sup \left\{ R_T : \nu_1, \dots, \nu_K \text{ in } \mathcal{Ber} \right\} \geq \Omega(\sqrt{TK})$$

References: Auer, Cesa-Bianchi, Freund, Schapire 2002

Bound achieved over $\mathcal{D} = \mathcal{P}([0, 1])$ and even in the **adversarial setting** by MOSS

Audibert and Bubeck, 2009; also Tsallis-INF strategies, Zimmert and Seldin, 2018

kl-UCB+ and KL-UCB-switch strategies achieve **simultaneously** optimal distribution-dependent and distribution-free regret bounds

Still based on upper confidence bounds; Ménard and Garivier, 2017; Garivier, Hadiji, Ménard, Stoltz, 2022